

Element A	Information	Other Details	Date Added
	Defining the problem (Elements A - C) We went to Ms. Weaver to see what she needed. She said that the shelves in the art showcase that hold the pottery are flimsy and dangerous. We established the weight, measurements, and durability required for the shelves to successfully complete the goal of creating new, more reliable shelves. Ideation, Selection, & Prototype design (Elements D - F) The shelves needed to be the right size, weight, and appearance in order to fit her needs. We decided on Sketch 5 originally, but we ended up iterating it and did not add the string. This was a good idea because the string would make it look tacky and not as professional. The prototype building (Element G) Our prototype was very successful. When building it we worked together and we used our time to work on making our shelf the perfect measurements for the metal supports. We made multiple iterations such as adding small blocks below the shelves to reduce sliding. What went well for us is our measuring. We made sure to make proper measurements so the shelves could fit perfectly into the showcase. The process of cutting and		

creating the shelves themselves was very good. The dimensions and the clean cut make the shelves look how we wanted them to. Something that did not go well was our time management. We really wanted to paint the shelves but we didn't have enough time to so they don't look as presentable as we wanted.

A lesson we learned is that planning is the most important part of the engineering design process. If you just hop into the building without planning you aren't gonna do nearly as well. But if you take the time to plan and be smart about it, you'll do much better. Planning allows for a more efficient building process and more successful result.

What do you think designers need to consider or do based on your experience?

They should check their measurements multiple times before cutting to ensure there are no errors and uneven cuts in order to create a better prototype/ final product. They should also make sure that they manage their time properly. The last thing you want is to be rushing on the last few days to get your project done. But if you manage your time well you can more comfortably get the project done. One last thing you should take into account is the materials and tools you are using to build, you always want to double check that you are using the proper blades to cut and you have the right materials.